

Notes on the newly developed atmTurbulentMoistureFluxHumidity boundary condition for moisture flux in OpenFOAM

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1 Boundary condition

The atmTurbulentMoistureFluxHumidity boundary condition (Ridwan, 2026) calculates the specific humidity at the boundary using the provided vertical moisture flux data. The flux data can be in various forms, such as latent heat flux (Section 1.1), kinematic moisture flux (Section 1.2), and moisture mass flux (Section 1.3).

1.1 Latent heat flux (W/m²)

The boundary condition expression that calculates the specific humidity at the boundary using the provided latent heat flux, Q_E , is:

$$\nabla q = \frac{Q_E}{\rho L_v \gamma} \quad (1)$$

where ρ is air density, L_v is the latent heat of vaporization, and γ is the mass diffusivity. At 20°C, $\rho = 1.2 \text{ kg/m}^3$, $L_v = 2.45 \times 10^6 \text{ J/kg}$. For a latent heat flux of $Q_E = 424.9 \text{ W/m}^2$, the boundary condition in OpenFOAM can be written as:

```
1 wall_BC
2 {
3     type                atmTurbulentMoistureFluxHumidity;
4     MoistureSource      flux;
5     gammaEff            gammaEff; // from shEqn.H line 8 of the
6     buoyantBoussinesqHumidityPimpleFoam solver
7     Lv0                 2940000; // effectively rho*Lv
8     value               uniform 0.00904; // initial guess, typically from IC
9     HumidityFlux        424.4; // Latent heat flux in W/m^2
10 }
```

1.2 Kinematic moisture flux (kg/kg m/s)

The specific humidity at the boundary can also be calculated using the provided kinematic moisture flux, Q_{kin} :

$$\nabla q = \frac{Q_{kin}}{\gamma} \quad (2)$$

where,

$$Q_{kin} = \frac{Q_E}{\rho L_v} \quad (3)$$

The same example mentioned in Section 1.1 can be written in OpenFOAM as:

```
1 wall_BC
2 {
3     type            atmTurbulentMoistureFluxHumidity;
4     MoistureSource  flux;
5     gammaEff        gammaEff; // from shEqn.H line 8 of the
6         buoyantBoussinesqHumidityPimpleFoam solver
7     Lv0             1; // because rho*Lv is already inside HumidityFlux
8         provided below
9     value            uniform 0.00904; // initial guess, typically from IC
10    HumidityFlux      1.4435E-04; //Kinematic moisture flux in kg/kg m/s
11 }
```

1.3 Moisture mass flux (kg/m²/s)

Moisture flux can be provided as moisture mass flux, Q_m , as well:

$$\nabla q = \frac{Q_m}{\rho\gamma} \quad (4)$$

where,

$$Q_m = \frac{Q_E}{L_v} \quad (5)$$

The same example mentioned in Section 1.1 can be written in OpenFOAM as:

```
1 wall_BC
2 {
3     type            atmTurbulentMoistureFluxHumidity;
4     MoistureSource  flux;
5     gammaEff        gammaEff; // from shEqn.H line 8 of the
6         buoyantBoussinesqHumidityPimpleFoam solver
7     Lv0             1.2; // density only as Lv is already inside HumidityFlux
8         provided below
9     value            uniform 0.00904; // initial guess, typically from IC
10    HumidityFlux      1.7322E-04; // Moisture mass flux in kg/m^2/s
11 }
```

References

Ridwan, T. (2026). buoyantboussinesqhumiditypimplefoam. GitHub repository, accessed 2026-06-04.